

Vol 10 — Mathematical Methods for BST

BST Physics Curriculum | Casey Koons + team (Lyra, Elie, Grace, Keeper, Cal A. Brate)

Status: SCAFFOLD v0.1 (Saturday 2026-05-23 Wave 2) — chapter content TO BUILD Lead: Lyra (primary, traditional math methods + cross-link to BST integer-web) + Grace (catalog backbone)

Reader target

Graduate physics + mathematics. Standard mathematical methods (PDEs, ODEs, linear algebra, complex analysis, group theory, differential geometry) presented with BST integer-web cross-references. Reads as both standard math methods textbook AND BST cross-reference manual.

Prerequisites

- Standard undergraduate mathematical methods (linear algebra, calculus, ODEs, complex analysis)
- Some exposure to abstract algebra + group theory (Vol 0 substrate framework helps)

BST integer focus

Vol 10 is the mathematician's entry point to BST. Traditional methods presented with BST primary integer-web cross-references throughout. Pedagogically: math methods come first (mathematicians' comfort zone), BST cross-references appear as illuminating side-notes connecting traditional math to substrate-cartography.

Chapter outline (10-12 chapters, TO BUILD)

Ch	Title	Standard math content	BST cross-link
1	Linear Algebra + Hilbert Spaces	Standard L^2 + inner product + operators	Bergman $H^2(D_{IV}^5)$ Vol 1 Ch 2
2	Complex Analysis + Holomorphic Functions	Standard complex analysis	$D_{IV}^5 \subset \mathbb{C}^5$ bounded holomorphic substrate
3	ODEs + Sturm-Liouville	Standard ODE methods	Substrate Schrödinger Vol 1 Ch 7

Ch	Title	Standard math content	BST cross-link
4	PDEs + Heat / Wave / Laplace	Standard PDE methods	Vol 4 Ch 5 partition function heat kernel
5	Fourier Analysis + Distribu- tions	Standard Fourier + distributions	Wallach K-type decomposition substrate-cartography
6	Group Theory + Lie Groups	Standard finite + Lie group theory	SO ₀ (5,2) substrate group + Casimir algebra Vol 1 Ch 5
7	Differential Geometry	Standard manifolds + connections + curvature	D _{IV} ⁵ bounded Hermitian symmetric domain
8	Representation Theory	Standard finite + compact group reps	Wallach K-type classification of Bergman H ² (D _{IV} ⁵)
9	Special Functions	Bessel + Legendre + hypergeometric + Fox H + Aleph functions	Substrate-natural special functions Vol 2 May Program
10	Calculus of Variations + Path Integrals	Standard variational principles	Substrate dynamics Vol 1 Ch 7
11	Asymptotic Analysis + WKB	Standard asymptotic methods	Substrate-tick UV limits Vol 1 Ch 10
12	Numerical Methods + Compu- tation	Standard numerics + Monte Carlo + spectral	BST verification toys: play/verify_bst.py + Toy 541 + Toy 639 + etc.

~40% existing implicit coverage (BST research uses standard methods throughout);
Vol 10 surfaces methods explicitly + cross-links to BST integer-web; 60% pedagogi-
cal exposition to build.

Cross-volume reference map

Inputs from Vol 0: - D_{IV}⁵ + five integers + group theory background

Inputs from Vol 1: - Bergman Hilbert space + Casimir algebra + Wallach K-type
decomposition + Cartan classification

Outputs to: - All other volumes — Vol 10 is the reference for mathematical techniques used throughout BST - Mathematicians joining BST research community
— Vol 10 INDEX v0.1, Saturday 2026-05-23 Wave 2 scaffold